

ASSIGNMENT – 15

ADVANCED PROGRAMMING

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SESSION-SPRING

YEAR-2024-28

- Without Mutex (Race Condition)

```
1  #include <stdio.h>
2  #include <windows.h>
3
4  #define NUM_THREADS 4
5  #define INCREMENTS 1000000
6
7  long long counter = 0;
8
9  DWORD WINAPI increment(LPVOID arg)
10 {
11     for(int i = 0; i < INCREMENTS; i++)
12     {
13         counter++;
14     }
15     return 0;
16 }
17
18 int main()
19 {
20     HANDLE threads[NUM_THREADS];
21
22     printf("Running WITHOUT mutex...\n");
23
24     for(int i = 0; i < NUM_THREADS; i++)
25     {
26         threads[i] = CreateThread(NULL, 0, increment, NULL, 0, NULL);
27     }
28
29     WaitForMultipleObjects(NUM_THREADS, threads, TRUE, INFINITE);
30
31     printf("Final Counter: %lld\n", counter);
32     printf("Expected Counter: %d\n", NUM_THREADS * INCREMENTS);
33
34     for(int i = 0; i < NUM_THREADS; i++)
35     {
36         CloseHandle(threads[i]);
37     }
```

```
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2  #include <windows.h>
3
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6
7  long long counter = 0;
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9  DWORD WINAPI increment(LPVOID arg)
10 {
11     for(int i = 0; i < INCREMENTS; i++)
12     {
13         counter++;
14     }
15     return 0;
16 }
17
18 int main()
19 {
20     HANDLE threads[NUM_THREADS];
21
22     printf("Running WITHOUT mutex...\n");
23
24     for(int i = 0; i < NUM_THREADS; i++)
25     {
26         threads[i] = CreateThread(NULL, 0, increment, NULL, 0, NULL);
27     }
28
29     WaitForMultipleObjects(NUM_THREADS, threads, TRUE, INFINITE);
30
31     printf("Final Counter: %lld\n", counter);
32     printf("Expected Counter: %d\n", NUM_THREADS * INCREMENTS);
33
34     for(int i = 0; i < NUM_THREADS; i++)
35     {
36         CloseHandle(threads[i]);
37     }
38
39     return 0;
40 }
```

- With Mutex (Correct Synchronization)

```
mutex.c > main()
1  #include <stdio.h>
2  #include <windows.h>
3
4  #define NUM_THREADS 4
5  #define INCREMENTS 1000000
6
7  long long counter = 0;
8  HANDLE mutex;
9
10 DWORD WINAPI increment(LPVOID arg)
11 {
12     for(int i = 0; i < INCREMENTS; i++)
13     {
14         WaitForSingleObject(mutex, INFINITE);
15         counter++;
16         ReleaseMutex(mutex);
17     }
18     return 0;
19 }
20
21 int main()
22 {
23     HANDLE threads[NUM_THREADS];
24
25     mutex = CreateMutex(NULL, FALSE, NULL);
26
27     printf("Running WITH mutex...\n");
28
29     for(int i = 0; i < NUM_THREADS; i++)
30     {
31         threads[i] = CreateThread(NULL, 0, increment, NULL, 0, NULL);
32     }
33
34     WaitForMultipleObjects(NUM_THREADS, threads, TRUE, INFINITE);
35
36     printf("Final Counter: %lld\n", counter);
37     printf("Expected Counter: %d\n", NUM_THREADS * INCREMENTS);
```

```
mutex.c > main()
33
34     WaitForMultipleObjects(NUM_THREADS, threads, TRUE, INFINITE);
35
36     printf("Final Counter: %lld\n", counter);
37     printf("Expected Counter: %d\n", NUM_THREADS * INCREMENTS);
38
39     for(int i = 0; i < NUM_THREADS; i++)
40     {
41         CloseHandle(threads[i]);
42     }
43
44     CloseHandle(mutex);
45
46     return 0;
47 }
```

OUTPUT-

- Without Mutex (Race Condition)

```
Active code page: 65001
```

```
C:\Users\VITU\OneDrive\Desktop\c>cd "c:\Users\VITU\OneDrive\Desktop\c\" && gcc race.c -o race && "c:\Users\VITU\OneDrive\Desktop\c\"race
Running WITHOUT mutex...
Final Counter: 2048053
Expected Counter: 4000000
```

```
c:\Users\VITU\OneDrive\Desktop\c>
```

- With Mutex (Correct Synchronization)

```
c:\Users\VITU\OneDrive\Desktop\c>cd "c:\Users\VITU\OneDrive\Desktop\c\" && gcc mutex.c -o mutex && "c:\Users\VITU\OneDrive\Desktop\c\"mutex
Running WITH mutex...
Final Counter: 4000000
Expected Counter: 4000000
```

```
c:\Users\VITU\OneDrive\Desktop\c>
```